Adaptation of Computer Use Agents to New Software with User Manuals

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Paper version: arXiv 2610.23012v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Maintaining strong performance on unseen applications remains challenging.*

Adapting a CUA to a new app usually needs **human-annotated** tasks and trajectories. **User manuals** already describe the procedures.

Motivation: three obstacles

1. **Hidden functions.** Menus, dialogs and shortcuts; long-horizon tasks rarely succeed.
2. **Costly data.** Executable tasks and trajectories need human annotation.
3. **Manual drift.** Versions and settings differ from the real GUI.

Evaluation setup

OSWorld · 5 apps | Slack · 30 own tasks | UI-TARS-1.5 7B | EvoCUA 8B | 50-step budget | 5 runs

- **All online** task success; no offline accuracy reported.
- **Not held out:** one policy per app, trained on that app's manual.

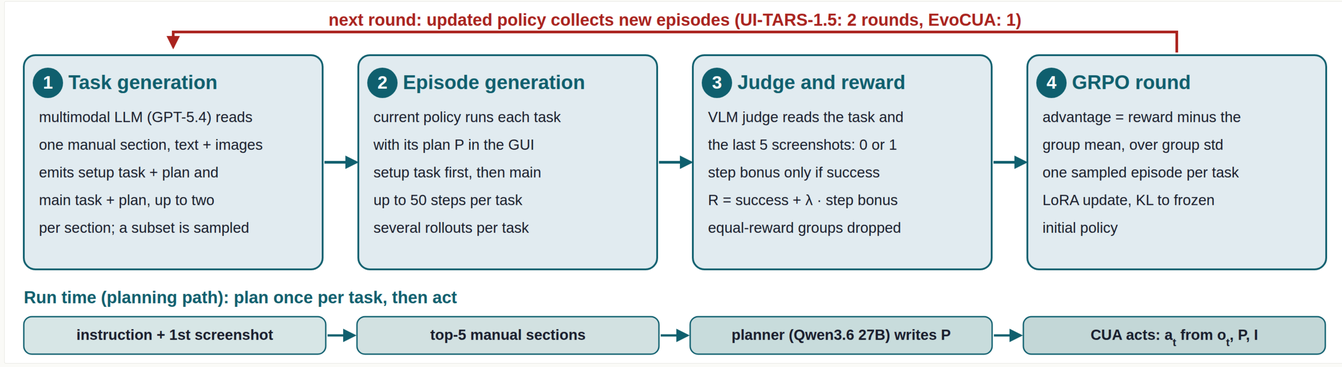
Our read (CUA Radar)

70	★★★	online
score of 100	band	evidence level

- **Reports:** online eval; 50-step budget; 5 runs; cost; ablation; judge vs rules; code.
- **Flags:** held-out apps only partial; judge reliability (AP4); ablation variants single-run.
- Pattern K28. Authors' numbers, not rerun.

Method: manual-retrieved plan at run time, GRPO on manual-derived tasks

- **Sees / acts:** screenshot in; click, drag, type, hotkey, scroll out.
- **Plan:** top-5 manual sections, then a Qwen3.6 27B planner writes one fixed plan.
- **Signal:** VLM judge on last 5 screenshots, 0/1 success plus step bonus; GRPO.



Our redraw (not the paper's figure): the manual-to-GRPO training loop and the run-time planning path.

$$R_i = R_i^{\text{success}} + \lambda R_i^{\text{step}}, \qquad R_i^{\text{step}} = \frac{T_{\max} - T_i}{T_{\max} - T_{\min}} \text{ if success, else } 0$$

Eqs. 2-3: success reward plus step-efficiency bonus, λ = 0.1

Key results: OSWorld task success (%), online

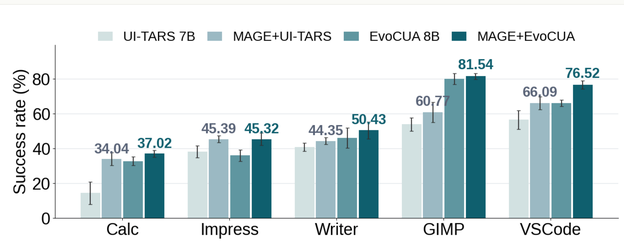


Table 1, redrawn: mean ± std, 5 runs, 50 steps; labels on MAGE bars.

Overall 36.83 → 47.31 (UI-TARS), 47.54 → 53.67 (EvoCUA)

SO WHAT → *Slack (30 own tasks): 26.67 → 49.33, 32.67 → 51.33.*

Ablation: OSWorld Calc success rate (%)

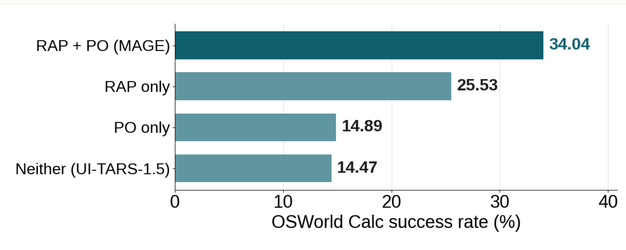


Table 3, redrawn: UI-TARS-1.5; baseline and MAGE are 5-run means, others one run.

SO WHAT → *RAP alone **+11.06 pp**, PO alone **+0.42 pp**, both **+19.57 pp**, as the paper states. PO helps mainly on top of RAP; the paper reads it as extra benefit.*

Headline Numbers

47.31%
OSWorld overall
UI-TARS-1.5 base: 36.83%

53.67%	EvoCUA + MAGE
49.33%	Slack
+19.57 pp	Calc, RAP + PO

The paper's own controls

CONTROL	MAGE / ALT.	VERSUS
Calc: manual 2021 / 2010	34.04 / 25.53	No plan 14.89
Calc: top-5 / random	34.04 / 27.66	Rank 6-10 14.89
Slack, 30 tasks	49.33	Base 26.67
Judge vs OSWorld rules	79.82%	Slack: none

Calc variants: 1 run each.

Takeaway

Both backbones gain on OSWorld; on Calc the manual plan carries most of the gain.

$37.83 API avg/app | 46.62 h, 3 H100 avg | judge agrees 79.82% | plans helpful 70.21%

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Paper | Code